JonathanMash

contact

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jonmash.com

education

Queen's University
2009 – 2013
M.Sc. in Electrical Eng.
Queen's Centre for Energy and Power Electronics Research
Thesis: Advanced Nonlinear Control Techniques for Wind Energy Conversions Systems
Course Avg: 92%

2004 – 2009
B.Sc. in Electrical Eng.
2nd/45 in Elec. Eng.
5th/576 in all Eng.
Final Year Avg: 93%

design tools

⋆Altium, Mentor Xpedition, Matlab, PSIM, SIWave, Visual Studio, ⋆Git/GitHub

programming

⋆Python, ⋆C/C++, Javascript, HTML5, CSS3, MySQL, Linux.

interests

Family, ⋆3D Printing, ⋆Woodworking, Electronics, Multirotor Drones, Solar Power Systems, Microcontrollers.

awards

2010 – *Ontario Graduate Scholarship*

2009 – *NSERC - Alexander Graham Bell Canada Graduate Scholarships*

skills

core: leadership, team & project management, problem solving, effective communication.
electronics: product development, system design, requirements, hardware/firmware interfaces.
hardware: pcb design, simulation, assembly & rework, testing & debugging, DfX.
mechanical: mechanical design, DfM/DfA, cross-domain hardware integration.
rapid prototyping: 3D printing, CNC machining, bench prototyping, design-for-iteration.
firmware: specifications, design, programming, testing & debugging, deployment.

experience

2023 – Present
BRINC Drones — Seattle, WA, USA
Director, Hardware Engineering
- Own all hardware development for the company's products – Electrical, Mechanical, and Test Engineering – from initial concept through production and in-field support.
- Led flagship products Lemur 2, Responder, and Responder Station from concept through production and beyond.
- Grew the engineering team at nearly 100% year-over-year headcount growth.
- Own and execute a multi-million dollar R&D budget; direct technical contributions to system definition, architecture, and key product specifications.
- Serve as the cross-functional steward between Hardware and Manufacturing, Software, Finance, Sales, Marketing, and Operations.

2020 – 2023
Amazon Leo (Kuiper) — Redmond, WA, USA
Hardware Development Manager - Avionics and Endpoints
- Recruited, trained, and managed a cross-functional team of over 14 electrical, system, and SI/PI engineers. Completed over 270 interviews at Amazon.
- Team lead and manager of the satellite Avionics & Bus Endpoints Team. Owner of flight computer, vehicle-wide communication bus, magnetometer, sun sensor, magnetorquers, propulsion control unit, solar array deployment unit, cameras, and more.
- Responsible for all aspects of the hardware design – architecture, requirements, development, testing, and qualification. Liaison between the Design Engineering team and the Systems, Reliability, Hardware Test, and Manufacturing Engineering teams.
- Maintained consistent and clear communication up the management chain. Owner of a multi-million dollar development budget.

2016 – 2020
Amazon Prime Air — Seattle, WA, USA
Hardware Development Manager - Aircraft Electronics
- Recruited, trained, and managed a cross-functional team of electrical, mechanical, system, and support engineers. Managed org-wide ECAD administration and design standards.
- Progressed from Hardware Development Engineer through Sr. Hardware Development Engineer to Manager – owned vehicle power distribution, sensor systems, ground station equipment, and RF link firmware along the way.

2010 – 2016
SPARQ Systems — Kingston, ON, Canada
Lead Product Developer – Grew the engineering group from one to six, delivering an embedded Linux IoT device with cloud-connected monitoring and control.